

PhD Applicant Research Task

Exploring LLM Agents for Scientific Software Engineering

1. Background

Large language models and coding agents are increasingly used for repository-level software engineering tasks, including issue understanding, fault localization, program repair, and testing. Scientific software introduces additional challenges because its correctness often depends on domain knowledge, numerical behavior, and scientific constraints.

This task is based on the SWE-bench Science benchmark and its accompanying paper:

SWE-bench Science: Can Coding Agents Resolve Engineering Tasks in Science?

Zhipeng Xu, Jiahao Lu, Yining Zheng, Yuxin Wang, Xipeng Qiu

[Paper](#) | [Dataset](#) | [Repository](#)

2. Task

Please study the paper and dataset, identify an interesting research problem, and design and implement your own solution. Possible directions include fault localization and repair.

Your work should:

- Formulate a clear and well-motivated research question;
- Propose and implement your own method;
- Compare the proposed method with at least one reasonable baseline;
- Conduct experiments on the full dataset or a well-motivated subset; and
- Present and critically analyze the results, including limitations where appropriate.

You are encouraged to explore freely and use the task to demonstrate your research ability, critical thinking, knowledge, technical skills, and capacity for independent investigation.

3. Deliverables

- An English research report in PDF format: no more than 4 pages of main content, plus up to 1 additional page for references;
- A repository link with commit history;
- A short README describing how to run the implementation and reproduce the main results; and
- Experimental results and the list of evaluated tasks.

The report should concisely describe the motivation, research question, proposed method, experimental setup, results, limitations, and conclusions. Please disclose the LLMs and AI-assisted development tools used.

4. Submission

Please send all materials by email to feifei.niu@tugraz.at no later than **16 September 2026, 23:59:59 Austria local time (CEST, UTC+2)**.

Email subject: **PhD Research Task Submission - [Full Name]**